Product backlog, sprint backlogs and Definition of Done

> **This is the planning record for releases R1 (v2.0) and R2 (v2.1). It is kept as history, not as the current plan.**
>
> The delivered plan is `../../docs/GroupMj Product-Backlog.xlsx` (`Mj01` – `Mj58`). That workbook is the activity-level record the brief asks for: every item has a first owner, a second owner, an estimate, acceptance conditions, a dependency and a status. The sprint workbooks are `../../docs/GroupMj Sprint-1-Backlog.xlsx` and `../../docs/GroupMj Sprint-2-Backlog.xlsx`.
>
> This document used `PB-nn` identifiers for a differently-scoped set of items and referred to owners as "team members A–E". Both caused the two plans to read as competing records of the same thing. The identifiers are retained below because they are the history of how the releases were planned, and because 08-test-plan.md originally traced against them. **§11 is the crosswalk** between them: every `PB-nn` here, the `Mj-nn` row in the delivered workbook that now carries it, and the test cases that exercise it. Traceability for new work runs against `Mj-nn` and nothing else.
>
> Where this document and the workbook disagree on a count, the workbook and the measured artefact win. The known disagreements are listed in §11.3.

> **Citations to files that are not in this tree.** An audit found 39 paths cited across 19 documents that no longer exist — the fourteen-pool model, the v1/v2 generators and the drawing scripts that went with them, and the Sprint 0 Node scaffold. Nothing was lost: all of it is in git history, and every citation is listed with what replaced it in 18-cited-files-no-longer-present.md.

Every item below is traced to an artefact. The model file, the forms and the worker sources are the source of truth; where a planning value (owner, estimate, sprint date) is not derivable from an artefact it is marked as a planning value and listed in the Assumptions section at the end.

Read with: `docs/01-gap-audit.md` (the measured starting point), `docs/02-status.md` (what is done and not done), `docs/03-test-record.md` (the execution record), `docs/04-modelling-decisions.md` (why the model is shaped as it is), `docs/05-v2.1-changes.md` (the second release), `docs/06-diagram-engineering.md` (the drawing: how it is routed, and what the third release changed).

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1. Product vision and scope statement

> Build one runnable BPMN collaboration of the hospital referral, treatment and administration pathway. Every hand-off between the named hospital teams is a real, correlated message. Clinical, administrative and financial decisions all sit with the correct role, and every modelled failure has an owner and an exit. The whole thing can be regenerated, deployed and demonstrated from the files in one shared workspace.

In scope for the product. The fifteen participants in the case study, and the whole pathway from referral receipt to follow-up. The scope also covers the nine hospital processes that carry executable work, the six outside participants as white-box pools of their own since v14 (`docs/11-white-box-external-participants.md`) at the boundary, and the

nine named exception conditions. Beyond the model itself: compensation for the two resources that are committed and can be released, the clinic-letter escalation ladder, the CAMUNDA Forms and their validation, the external workers that automate 35 job types, and the evidence that all of it runs.

Out of scope for the product (this module). Real integrations with any scheduling, correspondence, treatment, laboratory, imaging or payment provider; production identity, RBAC and persistence; a production database; and any deployment outside a local c8run cluster. See `docs/04-modelling-decisions.md` §7 and `workers/README.md` ("Simplifications and known gaps").

Module learning outcomes used in this document (legend only; the wording of the outcomes is not reproduced in the workspace, see Assumption A7):

Code	Short form used here
LO1	User-centred forms and data capture
LO2	A runnable, correct model and working automation
LO3	Project and test planning; configuration management
LO4	Business behaviour, exceptions and evidence of execution
LO5	Reflective, consistent, well-governed delivery

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2. Releases

Two releases exist, matching the two sprints and the two Sprint Reviews.

Release	Name	Files	What it adds	Backlog items
R1	**Initial Release** (v2.0)	`model/UFCEP6-0-3_Hospital_Patient_Pathway_v2.bpmn`, `model/forms/*.form`	Measured rebuild: new layout engine, lanes, collapsed subprocesses, exception branches with owners, compensation, capped retry, forms, workers, both run paths	PB-01 – PB-24
R2	**Second Release** (v2.1)	same filenames, regenerated	Visual compaction only: message-flow thinning, routine end-event merge, lane consolidation; nothing scored removed	PB-25 – PB-29

Definition of the Initial Release (R1). The Initial Release is accepted when: the v2.0 collaboration deploys to a local c8run 8.10.0-alpha5 cluster as 9 process definitions and 36 forms in one batch; the workers build and subscribe to every job type the model defines; the normal path and the unreadable-pack exception path each run to a stable state with `incidents: none`; and the layout measures zero sequence-flow overlaps and zero message-flow overlaps. All five conditions are evidenced in `docs/03-test-record.md` §C, §D, §E and §H.

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3. Priority scheme

MoSCoW is used, and within each band items are ranked 1 = first. Rank drives sprint selection; MoSCoW drives the release decision.

Band	Meaning for this product
Must	Without it the release is not accepted (see §2)
Should	Required for the 70%+ rubric band; carried as a committed stretch item
Could	Improves the portfolio; carried only if the Must and Should items are done
Won't (this release)	Explicitly deferred, with the reason recorded

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4. Product backlog

Estimates are in story points (SP) and ideal days (d). The workpackage owners for R1 and R2 were recorded as role letters: **A** model lead, **B** layout and visualisation lead, **C** workers lead, **D** forms and UX lead, **E** test and configuration lead. Every item has a first owner (accountable) and a second owner (reviewer and cover).

> **Role letters are history.** The R1/R2 letters map onto the four members and the initials the delivered workbook uses: **A** model lead — Lorrey (**L**); **B** layout and visualisation lead — Lorrey (**L**); **C** workers lead — Lorrey (**L**); **D** forms and UX lead — Damon (**D**); **E** test and configuration lead — Young (**Y**). Project management and the submission package were carried by Tommy (**T**), who does not appear as a letter here. The letters below are kept only so that the historical rows can still be read; new work is owned by initials.

4.1 Sprint 1 items: delivery of the Initial Release (R1)

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-01	1	Must	Measure v1.0 against every 70%+ rubric descriptor and list the gaps	LO3, LO5	5	1.5	E	A	Every descriptor in the gap audit has either a measurement or an artefact reference; every gap is numbered and actionable	—	Done	`docs/01-gap-audit.md`; `tools/analyse_layout.py`

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-02	2	Must	One BPMN collaboration with the 15 case-study participants; 9 executable hospital processes and 6 outside participants	LO2, LO4	8	2	A	E	15 `bpmn:participant` elements with the case-study names; 9 processes with `isExecutable="true"`; originally the 6 external bodies were black-box pools, superseded in v14: all 15 now carry a process and every outside participant is connected to another pool	PB-01	Done, extended in v14	model file participants; `docs/04-modelling-decisions.md` §1; `docs/11-white-box-external-participants.md`
PB-03	3	Must	Correlated message hand-off for every cross-pool transfer	LO2, LO4	8	2	A	C	68 message throw events and 10 catch events; every throw carries a `correlationKey` input mapping; message names are unique	PB-02	Done	model file; `docs/03-test-record.md` §B1; `docs/04-modelling-decisions.md` §2
PB-04	4	Must	Lanes inside pools that cover more than one desk	LO2	5	1.5	A	B	Every pool that covers more than one sub-role carries named lanes; participant names unchanged	PB-02	Done	14 lanes, e.g. `consultants_L0` / `consultants_L1`; `docs/03-test-record.md` §A7
PB-05	5	Must	Orthogonal layout engine: grid columns/rows, one track per flow, dedicated message channel	LO2, LO4	13	4	B	A	Sequence-flow overlap 0 pairs, message-flow overlap 0 pairs, 0 diagonal px; both `hospital-patient-pathway-v2.svg` and `.png` regenerate	PB-02	Done	`tools/layout_engine.py`; `docs/03-test-record.md` §A1–A3; `diagram/`
PB-06	6	Must	Contain repetitive administration in collapsed sub-processes	LO2	8	2	A	B	3 collapsed sub-processes, each with a `<bpmn:documentation>` note; their steps remain in the XML and executable	PB-05	Done	`SUB_Secretaries_Dispatch`, `SUB_CallHandling_Contact`, `SUB_Pathway_Report`; `docs/03-test-record.md` §A8
PB-07	7	Must	Every exception branch has an owner and an exit	LO4	13	4	A	C	All 9 `bpmn:error` codes reachable from 18 boundary events; no exception branch terminates without an owner task, a retry, a timer or an escalation	PB-03	Done	`docs/04-modelling-decisions.md` §4; `docs/03-test-record.md` §H1

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-08	8	Must	Clinic-letter delay timers and the three-rung escalation ladder	LO4	8	2	A	E	Boundary timers `P7D` and `R/PT168H` present; `letter.escalation-admin-manager` and `letter.escalation-higher-management` message chain present	PB-07	Done	model boundary events; `docs/03-test-record.md` §H1
PB-09	9	Must	A Camunda Form on every user task	LO1	8	2.5	D	A	39/39 user tasks carry `zeebe:formDefinition`; 36 distinct forms exist and every bound form is defined	PB-02	Done	`model/forms/` (36 `.form`); `docs/03-test-record.md` §H1, §H5d
PB-10	10	Must	Role-based candidate group on every user task	LO1, LO5	5	1.5	D	E	39/39 user tasks carry `zeebe:assignmentDefinition` with a candidate group matching the pool's role	PB-09	Done	`docs/03-test-record.md` §H1, §H5d
PB-11	11	Must	Validation rules on the forms	LO1	5	1.5	D	B	Required, length and numeric-range rules present on the fields that need them; long text is bounded	PB-09	Done	172 `validate` blocks across the 36 forms; `docs/01-gap-audit.md` §C2
PB-12	12	Should	Accessibility: field descriptions on every field, and grouping on long forms	LO1, LO5	5	1.5	D	E	Every field has a description; forms longer than ~8 fields are split by group heading	PB-09	**Partly done**: 71 descriptions across 230 fields; 0 `group` components emitted	form JSON census; `docs/01-gap-audit.md` §C3
PB-13	13	Must	One external worker per job type the model uses	LO2, LO4, LO5	21	6	C	A	Every job type in the model has a subscribing worker; build green; subscriptions observed on the live gateway	PB-02	Done	`workers/src/.../tasks/` (36 classes); `workers/README.md`; `docs/03-test-record.md` §E1–E2
PB-14	14	Must	Simulated external services with documented forced-failure switches	LO4	8	2.5	C	E	The 5 simulated job types can each be forced to succeed or fail without editing code, through `application.yaml` or an environment variable	PB-13	Done	`SimulatedFailureRegistry`; `workers/src/main/resources/application.yaml`; `workers/README.md`

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-15	15	Must	Duplicate suppression for bookings, charges and hand-offs	LO2, LO4	8	2	C	B	Repeating a booking returns the same reference; a duplicate charge is detected before payment; the correlation-key purpose tag prevents a duplicate hand-off	PB-03, PB-13	Done	`HospitalStore`; `workers/README.md`; `docs/04-modelling-decisions.md` §2
PB-16	16	Must	No storage or emission of complete card data	LO5	3	1	C	E	No payment worker reads or writes a card-number, CVV or expiry variable; only a payment reference and status are returned	PB-13	Done	`workers/README.md` job types 13 and 24 ("Never emits card data")
PB-17	17	Must	Audit-trail workers that do not overwrite clinical variables	LO4, LO5	5	1.5	C	A	Clinical and financial decisions produce an audit reference and timestamp; the audit workers refuse a blank `patientRef` and write no clinical variable back	PB-13	Done	`workers/README.md` job types 2 and 3
PB-18	18	Must	Compensation for the two committed resources	LO2, LO4	13	4	A	C	A compensation boundary event and `isForCompensation` handler on the appointment series and on the cycle booking; a compensation throw on the path that gives up; handlers typed `treatment.release-series` and `treatment.release-cycle-booking`	PB-07, PB-13	Done for modelling; not triggered live	model `TRT_Bnd_CompSeries`, `TRT_Bnd_CompCycle`, `TRT_Comp_ReleaseSeries`, `TRT_Comp_ReleaseCycle`; `docs/04-modelling-decisions.md` §5
PB-19	19	Must	Counted retry with a cap and an escalation for unavailable external capacity	LO4	8	2	A	C	A counted retry capped at 3; on exhaustion the provisional series is released and the case is handed to the pathway team, then retried in three days	PB-07	Done	`TRT_Auto_RecordRetry` (`treatment.record-capacity-retry`); `docs/04-modelling-decisions.md` §4
PB-20	20	Must	Deploy the model and all forms in one batch, repeatably	LO5	5	1.5	E	C	One command deploys 9 process definitions and 36 forms with no error, from a clean checkout, against `http://localhost:8080`	PB-05, PB-09	Done	`tools/deploy.sh`; `docs/03-test-record.md` §C1, §H2a
PB-21	21	Must	Normal-path run evidence	LO4	5	1.5	E	C	Eleven steps reached; `incidents: none` reported at the end of each stage; log retained	PB-20	Done	`screenshots/run-happy-path.log`

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-22	22	Must	Exception-path run evidence	LO4	5	1.5	E	A	`REFERRAL_PACK_UNREADABLE` is thrown by the worker, caught by `SEC_Bnd_DocumentsUnreadable`, and the case joins the missing-information loop with no incident	PB-20	Done	`screenshots/run-exception-path.log`
PB-23	23	Must	Prove every form renders	LO1, LO5	3	1	D	E	All 36 forms import and render in `@bpmn-io/form-js` with no unsupported field type	PB-09	Done	`screenshots/camunda-forms-rendered.png`; `docs/03-test-record.md` §F4
PB-24	24	Must	Written test record and modelling decisions document	LO3, LO5	8	2.5	E	A	Expected vs actual for every check run; every defect recorded; limitations stated before a reviewer finds them	PB-21, PB-22	Done	`docs/03-test-record.md`, `docs/04-modelling-decisions.md`

4.2 Sprint 2 items and deferred work

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-25	25	Must	Preservation audit as executable checks	LO3, LO5	8	2.5	E	B	Parsed checks over pools, user tasks, forms, candidate groups, job types, gateway types, error codes, timers and Camunda 7 attributes; all pass before a rebuild is accepted	PB-24	Done	`tools/verify_preservation.py` (62 checks, 62 passed); `docs/03-test-record.md` §H1
PB-26	26	Must	Thin the drawn message flows to one line per pool pair, keeping exception hand-offs	LO2	8	2.5	B	A	Message flows 66 → 31; dashed-line length 421,424 → 174,820 px; both run paths still clean afterwards	PB-25	Done	`docs/05-v2.1-changes.md` §1; `docs/03-test-record.md` §H3
PB-27	27	Must	Merge routine end events, never exception ends	LO2	5	1.5	B	E	End events 60 → 32; `_exception_ends()` keeps every exception end distinct	PB-26	Done	`docs/05-v2.1-changes.md` §2

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-28	28	Should	Consolidate single-step lanes into a neighbour lane of the same pool	LO2	3	1	B	A	Lanes 20 → 14; no participant renamed or removed	PB-26	Done	`docs/05-v2.1-changes.md` §3
PB-29	29	Must	Re-validate in the Modeler and export the diagram and per-pool sections	LO4, LO5	8	2.5	B	E	Camunda Modeler 5.51.0 reports 0 errors on Camunda 8.10 (alpha); SVG and PNG regenerate; 15 section crops refreshed	PB-26	Done	`screenshots/camunda-modeler-v2.1-status-bar.png`; `tools/render_diagram.sh`; `tools/export_sections.py`; `diagram/sections/`
PB-30	30	Should	Drive the three branches that are modelled but not run: declined payment, confirmation lost, three-month escalation	LO4	8	2.5	E	C	Each branch driven to its end state on the live engine, with a retained log and an Operate screenshot	PB-20	**Carried over**, not run	`docs/03-test-record.md` §D3, §G, §H6e; `docs/02-status.md` ("Not done")
PB-31	31	Should	Trigger compensation in a live run	LO4	5	1.5	C	E	A compensation handler completes inside a real process instance, visible in Operate	PB-18, PB-20	**Carried over**, not run	`docs/03-test-record.md` §D3
PB-32	32	Could	Fix the c8run Tasklist authentication so the forms draw in Tasklist	LO1, LO5	3	1.5	E	D	The Tasklist Task tab renders a v2.0 form for a claimed task	PB-20	**Carried over**: Tasklist gets `401` from `/v2/authentication/me`; environmental	`docs/03-test-record.md` §F5; `docs/02-status.md`
PB-33	33	Could	Reduce message start events to ≤ 2 per pool by changing the message protocol	LO2	13	4	A	C	Every hand-off publishes `Msg_<pool>` plus a `requestType` discriminator and each pool has one start event feeding a dispatch gateway	PB-03	Not started; deferred, see `docs/05-v2.1-changes.md` §4	arithmetic in `docs/05-v2.1-changes.md` §4
PB-34	34	Could	Crossing-reduction pass	LO2	8	2.5	B	A	Sequence-flow crossings reduced from 197 without reintroducing an overlap	PB-27	**Done in R3**: 94 crossings (–52%), 0 clipped lines, 0 overlaps held	`docs/06-diagram-engineering.md` §3.1, §3.4

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-35	35	Should	Readable terminals and hand-offs: one end event per terminating path, and dashed lines cut to the hand-offs that carry an outcome	LO2, LO4	5	1.5	B	D	Every end event has exactly one incoming sequence flow; no end event unreachable; every exception hand-off still drawn; no more than 15 message flows drawn	PB-27	**Done in R3**: 66 end events, none merged; 15 drawn message flows. v14 raises the message-flow ceiling to 22 to connect the six outside participants	`docs/06-diagram-engineering.md` §3.5, §3.6; `docs/11-white-box-external-participants.md`
PB-52	52	Won't (this release)	Replace the five simulated external services with real provider APIs	LO4	—	—	C	E	Not applicable: out of scope for a local portfolio release	PB-14	Won't (this release)	`workers/README.md` ("Simplifications and known gaps")

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5. Sprint backlogs

Sprint lengths, stand-up dates and review dates are planning values; no calendar record exists in the workspace (Assumption A3).

5.1 Sprint 1, to Sprint Review 1, delivering the Initial Release (R1)

Sprint Goal (agreed at planning).

> Replace the v1.0 drawing with a measured layout engine and close the two behavioural gaps the gap audit found (exception branches with an owner, and compensation for committed resources), then prove the whole collaboration runs on a live engine with no incidents.

Selection and control. All 24 Sprint 1 items entered the sprint. PB-01 to PB-05 came first, because every later item depends on the structure and the layout; the sprint was deliberately not expanded when the forms defect (§B of `docs/03-test-record.md`) was found. PB-12 was descoped during the sprint to descriptions only, and the grouping half is retained as unfinished.

Sprint 1 selection	Items	State at Sprint Review 1
Analysis and structure	PB-01, PB-02, PB-03, PB-04, PB-06	Completed
Drawing	PB-05	Completed
Behaviour	PB-07, PB-08, PB-18, PB-19	Completed for modelling; PB-18 not triggered live

Sprint 1 selection	Items	State at Sprint Review 1
Forms	PB-09, PB-10, PB-11, PB-12	PB-09/10/11 completed; PB-12 partly done
Workers	PB-13, PB-14, PB-15, PB-16, PB-17	Completed
Release and evidence	PB-20, PB-21, PB-22, PB-23	Completed
Documentation	PB-24	Completed

Completed at Sprint Review 1 (22 of 24). PB-01 – PB-11, PB-13 – PB-17, PB-18 (modelling only), PB-19 – PB-24. PB-12 was partly done; PB-18 was not triggered live.

Carried over out of Sprint 1.

Item	Carried because	Where it went
PB-12 (grouping half)	The field-description half was delivered; grouping was not implemented and no `group` component is emitted in any form	Not selected in Sprint 2; recorded as an open gap in `docs/08-test-plan.md` and `docs/10-sprint-reviews-and-feedback.md`
PB-18 (live trigger)	Compensation is modelled and the handlers exist; no live run exercised it	Re-framed as PB-31 in Sprint 2, still not run

5.2 Sprint 2, to Sprint Review 2, delivering the second release (R2)

Sprint Goal (agreed at planning).

> Reduce the drawn density of the shipped model without removing a single scored element, and re-prove the behaviour afterwards on the rebuilt file.

Selection and control. The committed set was PB-25 – PB-29, with PB-30 – PB-34 held as stretch items behind the Must items. PB-25 (the executable preservation audit) was built first so that every later change could be proved non-destructive. PB-30 – PB-34 were not started, and no committed item was dropped.

Sprint 2 selection	Items	State at Sprint Review 2
Committed	PB-25, PB-26, PB-27, PB-28, PB-29	Completed
Stretch	PB-30, PB-31, PB-32, PB-33, PB-34	None started
Deferred by decision	PB-35	Won't (this release)

Completed at Sprint Review 2 (5 of 5 committed). PB-25 – PB-29.

Carried over out of Sprint 2. PB-30, PB-31, PB-32, PB-33, PB-34, PB-35. Each carry-over keeps its MoSCoW band and its owner; the reason is recorded in the table in §4.2 and repeated in `docs/10-sprint-reviews-and-feedback.md`.

6. Definition of Done

An item is Done only when every applicable line below is true. The gates are the real commands in `tools/`; a claim without a gate result is not Done.

#	DoD condition	How it is proved	Applies to
DoD-1	The generator accepts the change: `python3 tools/build_v2.py` exits 0 with no check failure	Build output; the check refuses unbound forms, shared grid cells and mixed boolean/text comparisons	Model, forms
DoD-2	Layout gates hold: sequence-flow and message-flow overlaps are 0 pairs; no diagonals	`python3 tools/analyse_layout.py model/UFCEP6-0-3_Hospital_Patient_Pathway_v2.bpmn`	Drawing
DoD-3	Nothing scored was lost by a regeneration: 67 parsed checks pass	`python3 tools/verify_preservation.py`	Any model change after R1
DoD-4	The model parses clean in the Modeler's own parser: 0 warnings, 0 unresolved references, 0 dangling flows, 0 user tasks without a form or candidate group, 0 service tasks without a job type	`node tools/validate_model.js model/UFCEP6-0-3_Hospital_Patient_Pathway_v2.bpmn`	Model
DoD-5	The model opens in Camunda Modeler 5.51.0 with 0 errors on Camunda 8.10 (alpha)	Modeler status bar, screenshot retained	Release
DoD-6	The deployment is accepted: 9 process definitions and 36 forms in one batch	`bash tools/deploy.sh` output	Release
DoD-7	The workers build and subscribe to every model job type	`mvn -q -o clean package`; start-up subscription log	Workers
DoD-8	Both scripted paths reach their expected end state with `incidents: none`	`python3 tools/demo_scenario.py` and `--exception`; logs retained	Release
DoD-9	Every form imports and renders in `@bpmn-io/form-js`	Render census, `TOTAL=36 FAILED=0`; screenshot retained	Forms
DoD-10	Expected result, actual result and evidence file are written down	The test record is updated in the same change	Every item
DoD-11	The second owner has read the change and the evidence, and their name is against the item	Hand-over log in §7	Every item
DoD-12	Any limitation introduced is stated in `docs/04-modelling-decisions.md` §7 or `docs/02-status.md`	Document review at the sprint review	Every item that leaves a gap

7. How the two owners work, and the hand-over log

The pattern. The first owner is accountable: they make the change, run the gates that apply to it, and write the evidence line. The second owner reviews and covers. They reproduce at least one gate result themselves, read the evidence rather than the claim, and should be able to take the item over within one working session. No item is Done with only one owner's name on it (DoD-11). Where the two owners disagree, the item stays In progress and the disagreement goes to the stand-up; the first owner does not settle it alone. Cross-boundary items get a second owner from the other side of the boundary: the forms items are second-owned by the test lead, the worker items by the model lead. A claim of integration is then checked by someone who did not build the integration.

Cadence of hand-over. A hand-over is an explicit act, not an email: the artefact moves into the shared workspace, the gate output is pasted into the test record, and the row in the log below is closed in the same session. The log is the record of who held an item and when.

HO	From → To	Artefact handed over	Trigger	Accepted how	Status
HO-01	A → B	`tools/spec_v2.py` structure (pools, lanes, nodes, flows)	Structure frozen for Sprint 1	B builds the layout from the spec and returns the overlap measures	Closed
HO-02	B → A	Layout engine geometry and the analyse_layout output	After the first zero-overlap run	A confirms every node and flow is placed and no element moved between lanes	Closed
HO-03	A → C	Model job types and error codes	Before worker work	C confirms every service task type and every BPMN error has a worker behaviour	Closed (three v2.0 types added afterwards, logged as HO-08)
HO-04	D → E	36 `.form` files	Before the render census	E runs the form-js census and reports failures	Closed; the first run failed, see `docs/03-test-record.md` §F4
HO-05	E → D	Form render failure report (`date` field type)	Same day as HO-04	D fixes `tools/forms_spec.py` and the build re-emits the forms	Closed
HO-06	E → C	Deployment result (9 definitions, 36 forms)	Before the path runs	C runs the paths and returns the incident count	Closed
HO-07	C → E	Path logs	End of Sprint 1	E signs off the expected-vs-actual table	Closed
HO-08	A → C	Three v2.0 job types (`treatment.release-series` , `treatment.release-cycle-booking` , `treatment.record-capacity-retry`)	After compensation and capped retry were modelled	C confirms the three workers build and register	Closed
HO-09	B → E	v2.1 rebuilt model	Start of the compaction pass	E runs the 62-check preservation audit before anything else is accepted	Closed
HO-10	E → A	Preservation result and Modeler status bar	End of Sprint 2	A confirms the release note and the carried-over list	Closed

8. Agile working notes

Cadence. Two timeboxed sprints. Each sprint runs a graded stand-up at its midpoint and a graded Sprint Review at the end, so the demonstration is rehearsed once before it is graded. Planning is a single short session at the start of each sprint; the sprint backlog is frozen after it except for a removal, which must be agreed by both owners and recorded in §5.

Stand-up format (timeboxed, 10 minutes). Each of the five members answers in turn, with the artefact open rather than from memory: (1) which backlog item did you move, and which gate output proves it; (2) which item will you move next; (3) what is blocking you, and from whom. Blockers are written into the feedback action log in `docs/10-sprint-reviews-and-feedback.md` with a first and second owner before the stand-up ends. Status is only ever one of: Not started, In progress, Blocked, Done (per DoD), Carried over.

Review format (timeboxed). For each demonstrated item: the sprint goal, the increment actually shown, the acceptance evidence file, what was not finished, the decisions taken, and the limitations stated. The review runs the live path, not a slide. Anything not shown at the review is not counted as Done at that review even if it is finished afterwards.

Retrospective format (timeboxed, after each review). Three columns: what went well, what did not, and one improvement action. The action must name a first owner, a second owner and the sprint it lands in; an action with no owner is not recorded. The retrospectives are in `docs/10-sprint-reviews-and-feedback.md` §3 and §6.

Continuous integration of contributions. The workspace is integrated after every accepted item, not at the deadline. The generator is re-run, the layout gate and the preservation gate are executed, and the test record is updated in the same change. The deterministic build does the integrating: `build_v2.py` is documented as producing byte-identical output for the same inputs (`docs/05-v2.1-changes.md` §5), so an unexplained diff is a real change to review, not an accident of the tooling.

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9. Assumptions

1. **A1. Team identity.** Owners in §§4–5 are recorded as the role letters described in §4. The four members of the group are named in `../../docs/Contribution-Matrix.md` and are identified by their initials in the delivered workbook. *Corrected:* this assumption previously read "no real names appear in any artefact in the workspace, or in this document", which was untrue — the contribution matrix names all four members and gives each a share.
2. **A2. Estimates.** Story points, ideal days and ranks in this document are planning values authored for it. Estimates also exist in the delivered workbook, whose `Workload` sheet records an estimate for each of its 58 items (397 SP; 151 in Sprint 1 and 246 in Sprint 2). *Corrected:* this assumption previously read "no estimate exists in any artefact in the workspace", which the workbook disproves.
3. **A3. Dates.** Sprint length, stand-up dates and review dates are not recorded anywhere in the workspace. They are treated here as planning values for two timeboxed sprints.
4. **A4. Repository.** The workspace is a Git repository, and it is the version record: commits are releases and `v14.0-final`, `sr1-snapshot` and `sr2-snapshot` are the release and review snapshots. This document was first written when that was not true — it said "There is no `.git` directory … (checked)". The configuration-management plan it points at, `docs/09-risk-contingency-and-config-management.md` §4.1 and A1, carries the same correction.
5. **A5. Job-type counts.** The shipped v16.0 model defines 48 service tasks drawing on 35 distinct job types, and 4 elements throw `publish-message`. The worker sources contain 36 task classes producing 37 job-type subscriptions (35 model types, one alias `pathway.find-overdue-clinic-letters`, and `publish-message`). The v2.0-era

figures quoted in this document (47 service tasks, 68 throw events) describe the releases this document planned, not the shipped file. `workers/README.md` §4 carries the current catalogue and agrees with the counts above.

6. **A6. Diagram document (closed in R3).** The earlier `docs/06-visual-compaction.md` described a compaction that was never in the shipped model. In the third release that file was deleted and replaced by `docs/06-diagram-engineering.md`, which describes the drawing that is actually shipped and records the abandoned compaction as a rejected experiment with its measurements. The model file remains the source of truth; the drift is closed rather than carried.
7. **A7. Learning-outcome legend.** The full outcome wording is not reproduced in the workspace; the short forms in §1 are inferred from the descriptor/LO mapping used in `docs/01-gap-audit.md`.
8. **A8. Release naming.** "Initial Release" is R1 (v2.0) and the "second release" is R2 (v2.1). Both filenames are the same in the folder; the distinction is the content and the deployment version, not the path.

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11. Crosswalk: the `PB-nn` items here and the delivered `Mj-nn` plan

The group planned the work twice, in two records with different scopes and different identifier schemes. That was a mistake: it left the test plan tracing against identifiers that the delivered workbook does not contain. The workbook is now the plan of record. This section exists so the history stays readable and so no `PB-nn` reference elsewhere in the folder is a dead key.

Reading the table. `Mj` is the row in `../../docs/GroupMj Product-Backlog.xlsx` that carries the same work today. Where a `PB-nn` item describes an intermediate step — a v2.0/v2.1 planning release, or a measurement now recorded inside one of the delivered documents — it maps onto the item that owns that outcome.

11.1 Item crosswalk

`PB-nn`	What it planned	`Mj-nn` today	Note
PB-01	Gap audit against the 70%+ descriptors	Mj50	The measurement lives in `01-gap-audit.md`; Mj50 owns the plan
PB-02	One collaboration of the case-study participants	Mj15	Scope changed: v16.0 ships one hospital pool + 6 outside participants, not 15 processes
PB-03	Correlated message hand-off per cross-pool transfer	Mj15, Mj28	15 message flows on the shipped file
PB-04	Lanes inside multi-desk pools	Mj15	3 responsibility lanes holding 9 team lanes
PB-05	Orthogonal layout engine	Mj15	`06-diagram-engineering.md`
PB-06	Collapsed sub-processes for repetitive admin	Mj15	
PB-07	Every exception branch has an owner and an exit	Mj16, Mj21, Mj23	
PB-08	Letter delay timers and escalation ladder	Mj25	

`PB-nn`	What it planned	`Mj-nn` today	Note
PB-09	A form bound to every user task	Mj36–Mj44	38 user tasks, 37 forms
PB-10	Candidate group on every user task	Mj15, Mj46	
PB-11	Form validation	Mj45	
PB-12	Accessibility and field descriptions	Mj45	Delivered: 237/237 fields described, 15 groups
PB-13	A worker per job type	Mj29–Mj34	
PB-14	Simulated services and forced-failure switches	Mj29–Mj33, Mj53	
PB-15	Duplicate suppression	Mj22, Mj33	
PB-16	No card data in any variable	Mj41, Mj48	
PB-17	Audit workers that do not overwrite	Mj47	
PB-18	Compensation for committed resources	Mj20	
PB-19	Capped external-capacity retry	Mj20, Mj33	
PB-20	Deployment configuration	Mj28, Mj58	
PB-21	Normal-path run evidence	Mj52, Mj56	
PB-22	Exception-path run evidence	Mj52	
PB-23	Form rendering census	Mj44, Mj53	
PB-24	Test record and modelling decisions	Mj50, Mj51, Mj55	
PB-25	Preservation audit	Mj55	The tool that ran it has been removed; see §11.3
PB-26	Message-flow thinning	Mj15	
PB-27	One end event per terminating path	Mj15	
PB-28	Lane consolidation	Mj15	
PB-29	Modeler validation and export	Mj54, Mj57	
PB-30	Driving the undriven branches	Mj52	Partly: 8 scenarios on v16.0; the worker failure paths remain **not run**
PB-31	Live compensation trigger	Mj20, Mj52	Achieved on v16.0: `screenshots/run-v16-compensation.log`
PB-32	Tasklist form display	Mj45, Mj54	Achieved on v16.0: `screenshots/v16/tasklist-form-rendered.png`
PB-33	Start-event reduction	Mj15	Deferred, with the arithmetic recorded in `05-v2.1-changes.md` §4

`PB-nn`	What it planned	`Mj-nn` today	Note
PB-34	Crossing-reduction pass	Mj15, Mj57	Measured in `screenshots/v16/convergence-audit.txt`
PB-35	Readable terminals and hand-offs	Mj15	
PB-36	Rebuild the model as one hospital pool	Mj15	Achieved; `17-single-pool-operational-model.md`
PB-37	Deploy and drive every scenario	Mj52	8 scenarios COMPLETED
PB-38	Build checks on the form↔model contract	Mj28, Mj53	`tools/spec_v3.py`
PB-39	Accessibility pass, contrast measured	Mj45, Mj54	
PB-40	Stand-up and sprint-review records	Mj05, Mj09, Mj06, Mj10	Partly: reconstructed, not contemporaneous
PB-41	Rebuild the worker jar from shipped source	Mj35, Mj58	
PB-42	Repair the two results their evidence contradicted	Mj46, Mj55	
PB-43	Index citations to files no longer in the tree	Mj58	`18-cited-files-no-longer-present.md`
PB-44	Correct the falsely-Done backlog rows	Mj45, Mj46, Mj47	Those three now read "Partly done"
PB-45	Worker unit tests and an engine integration test	Mj35	49 tests; the 35 domain `execute()` methods still have none
PB-46	A recorded demonstration with a person filling forms	Mj56, Mj57	**Not done:** no person has completed a form through Tasklist
PB-47	Worker failure paths on v16.0	Mj55	**Not done**
PB-48	Grouped forms submitted fields one level deep	Mj45	Fixed and guarded
PB-49	A textarea read as a list	Mj39, Mj45	Fixed and guarded
PB-50	A hidden required field submitted empty	Mj39, Mj45	Fixed and guarded
PB-51	Empty document checklist	Mj36	Fixed
PB-52	Real provider APIs instead of simulated services	—	Out of scope, stated as a limitation

11.2 Test cases by delivered item

`08-test-plan.md` §8 traces against `Mj-nn`. The `PB-nn` keys below are the historical ones; the test-case identifiers are unchanged.

`Mj-nn`	`PB-nn`	Test cases
Mj15	PB-02, PB-03, PB-04, PB-05, PB-06, PB-26, PB-27, PB-28, PB-33, PB-34, PB-35, PB-36	TC-BPMN-01…-09, -13, -15, -18, -20, -21, -22, -24, -25
Mj16	PB-07	TC-BPMN-16, TC-WRK-05, TC-E2E-02

`Mj-nn`	`PB-nn`	**Test cases**
Mj20	PB-18, PB-19, PB-31	TC-BPMN-16, TC-E2E-06
Mj25	PB-08	TC-E2E-05
Mj28	PB-20, PB-38	TC-DEP-01…-04
Mj33	PB-15, PB-19	TC-WRK-07, -08, -09
Mj36–Mj44	PB-09, PB-23	TC-BPMN-19, TC-FRM-01, TC-FRM-02, TC-FRM-06
Mj41	PB-16	TC-NFR-01, TC-NFR-08
Mj45	PB-11, PB-12, PB-32	TC-FRM-03, -04, -05, -07, -08
Mj46	PB-10	TC-BPMN-10, TC-NFR-02
Mj47	PB-17	TC-NFR-02, TC-NFR-03
Mj50, Mj51	PB-01, PB-24	TC-BPMN-11, TC-E2E-08
Mj52	PB-21, PB-22, PB-30	TC-E2E-01…-05
Mj54	PB-29, PB-32	TC-BPMN-10, TC-BPMN-11, TC-NFR-06
Mj55	PB-25, PB-45	TC-BPMN-12, -13, -14, -17, -18
Mj57	PB-29, PB-34	TC-BPMN-04, -09, -22

11.3 Known disagreements between this document and the delivered artefacts

Corrections are applied here rather than carried, so that a reader does not have to work out which record to trust:

This document said	**The delivered artefact says**	**Where**
Team of five, owners A–E	Four members, initials T/D/Y/L	`../../docs/Contribution-Matrix.md`; the workbook
No real names in any artefact	All four are named with shares	Corrected in A1 above
No estimate in any artefact	397 SP over 58 items	Corrected in A2 above
47 service tasks, 68 throw events	48 service tasks, 4 throw events	Corrected in A5 above
9 executable processes, 36 forms	1 executable process, 37 forms	§2; `README.md`
PB-35 used for two different items	—	The deferred integration item is now PB-52
Traceability keyed to `PB-nn`	The workbook uses `Mj-nn`	§11.2; `08-test-plan.md` §8

v16.0 rows (the R5 change set)

> **Read §11 first.** These rows are the release history of the v16.0 rebuild. Their `PB-nn` identifiers crosswalk to the delivered workbook in §11.1. The columns are the same as §4.1, so an item here can be read the same way.

The audit that started this increment found the v16.0 model absent from version control, a regression of DEF-02, two results contradicted by their own cited evidence and a shipped jar built from an earlier release; PB-36 to PB-44 are the response, and PB-48 to PB-51 are the defects a person found afterwards by using Tasklist.

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-36	36	Must	Rebuild the operational model as one hospital pool with responsibility lanes	LO2, LO4	21	6	A	B	One executable process; 3 responsibility lanes holding 9 team lanes; 7 participants; every user task bound to a form and a candidate group; every service task bound to a job type	PB-07	**Done**	`model/UFCEP6-0-3_Hospital_Patient_Pathway_v16.bpmn`; `docs/17-single-pool-operational-model.md`
PB-37	37	Must	Deploy the rebuilt model and drive every scenario end to end	LO2, LO4, LO5	13	4	E	A	Each scenario ends `COMPLETED` with no incident left open, driven only by the fields of the bound forms	PB-36	**Done for the eight scenarios in `docs/03` K2; NOT RUN for the worker failure paths in K3**	`screenshots/run-v16-*.log`; `screenshots/v16/engine-record.txt`
PB-38	38	Must	Build checks that refuse a model whose conditions do not match its forms	LO2, LO5	8	2.5	A	C	A generator that emits a model whose gateway conditions disagree with the bound form fails the build, with the offending condition named	PB-36	**Done**	`tools/spec_v3.py` (`check_conditions`, `check_gateway_variables`, `check_gateway_sources`, `check_option_values`, `check_form_worker_types`)
PB-39	39	Should	Accessibility pass over every form	LO1, LO5	8	2.5	D	E	Every field carries a description; every control carries a human label; every choice offers words; every long form is grouped **and the contrast of the rendered form is measured**	PB-36	**Done**	`tools/audit_accessibility.py`; `tools/audit_rendered_contrast.py`; `screenshots/v16/accessibility-report.json`; `screenshots/v16/tasklist-contrast.json`; `docs/20-accessibility-and-the-forms.md`

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-40	40	Must	Stand-up and sprint-review records	LO3, LO5	5	1.5	E	A	Both stand-ups recorded with progress, evidence, next action, blocker and handover; both reviews recorded with the increment, the decisions and what was carried	PB-37	**Partly done** — reconstructed from the commit log; no contemporaneous notes existed	`docs/19-standups.md` ; `docs/10-sprint-reviews-and-feedback.md` §R5
PB-41	41	Must	Rebuild the worker jar from the shipped source	LO4, LO5	3	1	C	E	The jar in the tree contains `hospital/db/` , `DateRuleValidator` , `JobSecurity` , the H2 driver and the current `date-rules.json` , and the documented run command starts the shipped code	PB-36	**Done**	`workers/target/hospital-external-workers-1.0.0.jar` (32,058,075 bytes)
PB-42	42	Must	Repair the two results whose cited evidence contradicted them	LO5	5	1.5	E	A	The RBAC capture that reads all zeros is headed as such and its citations repointed; the compensation claim is corrected in place and compensation is modelled and run	PB-37	**Done**	`screenshots/v15/users-roles-authorizations.txt` ; `docs/14-feedback-response.md` §1; `docs/16-acceptance-criteria.md` §6
PB-43	43	Should	Index every citation to a file that is no longer in the tree	LO5	3	1	E	B	Every cited path that no longer exists is listed with what replaced it, and the affected documents point at the index	—	**Done**	`docs/18-cited-files-no-longer-present.md`
PB-44	44	Must	Correct the backlog rows the team had recorded as falsely Done	LO3, LO5	2	0.5	E	A	Mj45, Mj46 and Mj47 no longer read Done	—	**Done**	`docs/GroupMj Product-Backlog.xlsx` ; `docs/16-acceptance-criteria.md` §5
PB-45	45	Must	Worker-layer unit tests and an engine integration test	LO5	13	4	C	E	`AbstractHospitalWorker` 's complete / BPMN-error / fail-with-retries mapping is covered, `JobContext` validation is covered, and at least one `execute()` is covered end to end against an engine	PB-41	**Partly done** — 49 tests, of which `JobContext` (13) and `AbstractHospitalWorker` (6) are new, so the base class is covered; no engine integration test, and the 36 `execute()` methods still have none	`workers/src/test/` ; `TestJob.java` and `RecordingJobClient.java` for the doubles; `docs/03` §K6

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-46	46	Should	A recorded demonstration with a person filling the forms in Tasklist	LO1, LO4, LO5	5	1.5	E	D	A recording of a case driven through Tasklist by hand, with the service tasks visibly completing behind it	PB-37, PB-39	**Partly done** — the form is now opened, measured and photographed in the real Tasklist (`docs/20`), but no person has completed one. A scripted attempt to fill the pack-check form could not satisfy it: `Complete Task` stayed disabled, which is the clearest statement of why this one needs a person	`screenshots/v16/tasklist-form-rendered.png` ; `screenshots/v16/tasklist-contrast.json` ; `docs/20`
PB-47	47	Must	Worker failure paths on v16	LO4, LO5	8	2.5	E	C	Each of the nine error codes driven on the shipped model, with a retained log and the incident count	PB-37	**Not done**	`docs/03-test-record.md` §K3
PB-48	48	Must	Grouped forms submitted their fields one level deep, so no worker could read them	LO1, LO2, LO5	5	1.5	D	C	No form field is submitted under a group prefix; the grouping stays for the reader	PB-39	**Done** — found by a person using Tasklist; every automated run filled the forms over the API and never rendered one	`tools/fix_form_group_paths.py` ; guard `forms_inventory.py` `KEY_NESTED_BY_GROUP_PATH` ; `docs/20`
PB-49	49	Must	A textarea read as a list broke the funding approval	LO2, LO5	3	1	C	D	A free-text answer is accepted wherever a limitation is recorded, and no worker reads a list from a field that submits a string	PB-41	**Done**	`JobContext.stringListOrText` ; guard `spec_v3.check_form_worker_types` ; `docs/20`
PB-50	50	Must	A hidden required field submitted empty and failed two steps later	LO1, LO2, LO5	3	1	D	C	No conditional compares a string-valued field against an unquoted boolean; a required field that is hidden is reported	PB-39	**Done**	`model/forms/record-consent-and-treatment-request.form` ; guard `forms_inventory.py` `CONDITIONAL_TYPE_MISMATCH` ; `docs/20`

ID	Rk	Pri	Item	Why (LO)	SP	d	1st	2nd	Acceptance criteria	Depends on	Status	Evidence
PB-51	51	Must	The document checklist could be submitted empty, sending the case round the missing-information loop	LO1, LO4	2	0.5	D	E	The referral pack check cannot be submitted with no documents stated	PB-48	**Done**	`model/forms/check-referral-pack.form` (`supportingDocuments` required)